Pharmacy Management System - Project Scheduling

Practical 3: Scheduling Project

Aim:

To schedule tasks and resources for a Pharmacy Management System project using a Gantt chart to ensure efficient project planning and execution.

Introduction:

A Pharmacy Management System (PMS) automates various processes involved in managing a pharmacy, such as inventory management, sales, billing, and customer records. Efficient project scheduling is essential to manage tasks, allocate resources, and meet deadlines effectively. This project involves identifying tasks, assigning resources, and developing a project schedule using a Gantt chart.

Objective:

- **To identify key tasks involved in developing a Pharmacy Management System.**
- **To allocate resources effectively for each task.**
- **To create a project schedule using a Gantt chart.**
- **To monitor and manage the project timeline for successful completion.**

Theory:

Project scheduling is a fundamental process in project management that ensures tasks are completed on time and within budget. It involves defining tasks, estimating durations, assigning resources, and sequencing activities. The Work Breakdown Structure (WBS) is commonly used to break down the project into smaller, manageable tasks.

The Gantt chart is a widely used scheduling tool that visually represents tasks along a timeline, showing their start and end dates, dependencies, and progress. Critical Path Method (CPM) helps identify the sequence of dependent tasks that determine the project's duration.

Detailed Key Steps in Project Scheduling:

1. **Task Identification:**

- **Identify and list all tasks required to complete the project.**
- **Break down tasks into smaller, manageable components.**
- **Categorize tasks based on project phases.**

2. **Resource Allocation:**
    - **Identify the resources needed (e.g., personnel, equipment, materials).**
    - **Assign tasks to team members based on skill sets.**
    - **Ensure resource availability and avoid over-allocation.**
3. **Timeline Estimation:**
    - **Estimate the duration for each task using historical data or expert judgment.**
    - **Use estimation techniques like Three-Point Estimation (Optimistic, Pessimistic, and Most Likely estimates).**
    - **Define task start and end dates.**
4. **Dependency Identification:**
    - **Identify relationships between tasks (e.g., finish-to-start, start-to-start).**
    - **Sequence tasks logically to maintain workflow.**
    - **Recognize critical tasks that affect the project timeline.**
5. **Monitoring and Adjustment:**
    - **Track progress using project management tools.**
    - **Update schedules based on performance data.**
    - **Implement corrective actions if delays or issues arise.**

Gantt Chart for Pharmacy Management System:

Task	Start Date	End Date	Duration (Days)	Assigned Resources
Requirement Gathering	01/03/2025	05/03/2025	5	Project Manager, BA
System Design	06/03/2025	12/03/2025	7	Developers, BA
Development	13/03/2025	30/03/2025	18	Developers
Testing	01/04/2025	10/04/2025	10	Testers
Deployment	11/04/2025	15/04/2025	5	IT Support
Maintenance	16/04/2025	Ongoing	-	IT Support

Case Study: Pharmacy Management System

A pharmacy management system aims to automate inventory management, sales tracking, and billing. The development process includes tasks like requirement gathering, system design, development, testing, deployment, and maintenance. The scheduling ensured timely completion and optimal resource utilization.

Conclusion:

The scheduling process for the Pharmacy Management System was streamlined by identifying essential tasks, assigning resources, and creating a Gantt chart. This approach provides a clear overview of the project timeline and facilitates effective team collaboration, ensuring the project's successful completion within the set timeframe.